

Graphics Card Market Analysis - Web Scraping Project

Project Overview

In this project, you will create a web scraper to collect and analyze graphics card data from Newegg.com's GPU section. You'll use Python, BeautifulSoup, and other relevant libraries to extract key information about graphics cards currently available in the market.

Learning Objectives

- Implement web scraping techniques using Python and BeautifulSoup
- Handle pagination in web scraping
- Process and structure collected data
- Practice working with real-world data
- Learn to respect website scraping policies and implement delays

Part 1: Web scrapping

Required Data to Extract

For each graphics card listing, you must collect:

1. Product Image URL
2. Product Title (e.g., "MSI Ventus GeForce RTX 3060 12GB")
3. Product Details:
 - Manufacturer
 - Model Number
 - Rating (number of stars)
4. Pricing Information:
 - Current Price
 - Original Price (if available)
 - Shipping Information

 Screenshot

Project Requirements

Technical Requirements

1. Use Python 3.x
2. Required libraries:
 - requests
 - beautifulsoup4
 - pandas
 - time (for implementing delays)

Functional Requirements

1. Implement proper error handling
2. Include appropriate delays between requests (minimum 2 seconds)
3. Save the scraped data to a CSV file with appropriate column headers
4. Document your code with comments
5. Create a summary of the collected data (e.g., average price, price range)

Part 2: Data Analysis

Required Analysis

1. Price Analysis
 - Calculate average price by brand
 - Identify price ranges and market segments
 - Analyze price-to-memory ratio
 - Compare prices across different GPU series (RTX 3000 vs 4000)
 - Create price distribution visualizations
2. Brand Analysis
 - Market share by brand
 - Average ratings by brand
 - Price comparison across brands
 - Create brand comparison visualizations
3. Memory Configuration Analysis
 - Distribution of memory sizes
 - Price correlation with memory size
 - Popular memory configurations by brand
4. Market Segmentation
 - Entry-level vs Mid-range vs High-end segments
 - Price brackets and their features

Deliverables

1. Python script containing the web scraper
2. CSV file with the collected data
3. README file explaining:
 - Project Description
 - Features
4. Short report analyzing the collected data

Important Notes

- Always check and respect the website's robots.txt file
- Implement appropriate delays between requests to avoid overwhelming the server
- Handle your requests responsibly and ethically
- Be prepared to handle cases where data might be missing or in an unexpected format

```
In [17]: # Import chrome webdriver to access chrome options
# Import chrome webdriver to access chrome options
from webdriver_manager.chrome import ChromeDriverManager
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.common.by import By
import pandas as pd
import time
import selenium
from selenium import webdriver
```

```
In [28]: url= 'https://www.newegg.com/GPUs-Video-Graphics-Cards/SubCategory/ID-48'
service = Service(ChromeDriverManager().install())
driver = webdriver.Chrome(service=service)
driver.get(url)
time.sleep(3)
cards= driver.find_elements(By.CSS_SELECTOR , 'div.item-cell')
data =[]
#Loop through each card and extract fields
for card in cards:
    ##Field 1
    try:
        image_url = card.find_element(By.CSS_SELECTOR , 'a.item-img img').get_attri
    except:
        image_url = None
    ##Field 2 Rating
    try:
        rating = card.find_element(By.CSS_SELECTOR , 'a.item-rating').get_attribute
    except:
        rating= None
    try :
        review = card.find_element(By.CSS_SELECTOR , 'span.item-rating-num').text.s
    except:
        review = None
    #Field 3 : Manufacturer (title attribute, fallback alt)
    try:
        brand_img=card.find_element(By.CSS_SELECTOR , 'a.item-brand img')
```

```

        brand = brand_img.get_attribute('title') or brand_img.get_attribute("alt")
    except:
        brand = None
#Field 4 Product Title
    try:
        title= card.find_element(By.CSS_SELECTOR , 'a.item-title').text
    except:
        title = None
## Field 5: Model number (split text after "Model #:")
    try:
        features_text = card.find_element(By.CSS_SELECTOR , 'ul.item-features').text
        model =features_text.split("Model #:")[-1].strip() if "Model #:" in features_text
    except:
        model = None
## Field 6 Current Price (dollars + cents combined)
    try:
        price_el = card.find_element(By.CSS_SELECTOR , 'li.price-current')
        dollars =price_el.find_element(By.TAG_NAME , "strong")
        cents=price_el.find_element(By.TAG_NAME , "sup")
        price =f"{dollars.text}{cents.text}"
    except:
        price = None
## Field 7: shipping info
    try:
        shipping = card.find_element(By.CSS_SELECTOR, 'li.price-ship').text
    except:
        shipping = None

## Save this card's data (runs for every card, not just failures)
    data.append({
        "Title": title, "Brand": brand, "Rating": rating,
        "Review_Count": review, "Model": model, "Price": price,
        "Shipping": shipping, "Image_URL": image_url
    })

df = pd.DataFrame(data)
print(df)
df.to_csv("newegg_products.csv", index=False)
driver.quit()

```

	Title	Brand \
0	SAPPHIRE PULSE Radeon RX 9060 XT 16GB GDDR6 PC...	Sapphire Tech
1	ASRock Challenger Radeon RX 9060 XT 16GB GDDR6...	ASRock
2	GIGABYTE Gaming GeForce RTX 5070 Ti 16GB GDDR7...	GIGABYTE
3	GIGABYTE Gaming GeForce RTX 5080 16GB GDDR7 PC...	GIGABYTE
4	ASRock Challenger Radeon RX 9070 XT 16GB GDDR6...	ASRock
5	GIGABYTE Gaming GeForce RTX 5070 12GB GDDR7 PC...	GIGABYTE
6	ASRock Challenger Radeon RX 9070 16GB GDDR6 PC...	ASRock
7	SAPPHIRE PULSE Radeon RX 9070 XT 16GB GDDR6 PC...	Sapphire Tech
8	GIGABYTE Gaming Radeon RX 9070 XT 16GB GDDR6 P...	GIGABYTE
9	ASRock Challenger Arc B580 12GB GDDR6 PCI Expr...	ASRock
10	ASRock Challenger Radeon RX 7600 8GB GDDR6 PCI...	ASRock
11	GIGABYTE WindForce GeForce RTX 5070 Ti 16GB GD...	GIGABYTE
12	GIGABYTE Gaming Radeon RX 9070 GRE 12GB GDDR6 ...	GIGABYTE
13	MSI Ventus GeForce RTX 5060 Ti 8GB GDDR7 PCI E...	MSI
14	SAPPHIRE NITRO+ Radeon RX 9070 XT 16GB GDDR6 P...	Sapphire Tech
15	XFX Quicksilver AMD Radeon RX 9070 XT Gaming E...	XFX
16	MSI FRIEREN EDITION GeForce RTX 5070 Ti 16GB G...	MSI
17	ASRock Radeon RX 7900 XTX 24GB GDDR6 PCI Expre...	ASRock
18	ONIX LUMI Arc B580 12GB GDDR6 PCI Express 4.0 ...	None
19	MSI Ventus GeForce RTX 5060 Ti 16GB GDDR7 PCI ...	MSI
20	GIGABYTE WINDFORCE GeForce RTX 5080 16GB GDDR7...	GIGABYTE
21	MSI Ventus GeForce RTX 5070 Ti 16GB GDDR7 PCI ...	MSI
22	XFX Swift AMD Radeon RX 9060 XT OC Gaming Edit...	XFX
23	GIGABYTE Eagle GeForce RTX 5070 12GB GDDR7 PCI...	GIGABYTE
24	MSI SHADOW GeForce RTX 5080 16GB GDDR7 PCI Exp...	MSI
25	XFX Mercury Radeon RX 9070 XT OC 16GB 256-Bit ...	XFX
26	GIGABYTE Gaming Radeon RX 9060 XT 16GB GDDR6 P...	GIGABYTE
27	GIGABYTE Eagle GeForce RTX 5070 Ti 16GB GDDR7 ...	GIGABYTE
28	ASUS ROG Astral GeForce RTX 5090 32GB GDDR7 OC...	ASUS
29	ASRock Steel Legend Radeon RX 9070 XT 16GB GDD...	ASRock
30	GIGABYTE WindForce GeForce RTX 5070 12GB GDDR7...	GIGABYTE
31	GIGABYTE Gaming Radeon RX 9060 XT 16GB GDDR6 P...	GIGABYTE
32	MSI SHADOW GeForce RTX 5070 12GB GDDR7 PCI Exp...	MSI
33	GIGABYTE Gaming GeForce RTX 5090 32GB GDDR7 PC...	GIGABYTE
34	MSI Gaming GeForce RTX 5090 32GB GDDR7 PCI Exp...	MSI
35	ASUS PRIME GeForce RTX 5070 Ti 16GB 256-Bit GD...	ASUS

	Rating	Review_Count	Model	Price \
0	Rating + 4.7	124	11350-03-20G	519.99
1	Rating + 4.7	109	RX9060XT CL 16GO	469.99
2	Rating + 4.8	241	GV-N507TGAMING OC-16GD	1,087.99
3	Rating + 4.6	300	GV-N5080GAMING OC-16GD	1,699.99
4	Rating + 4.6	139	RX9070XT CL 16G	719.99
5	Rating + 4.3	142	GV-N5070GAMING OC-12GD	839.99
6	Rating + 4.5	74	RX9070 CL 16G	649.99
7	Rating + 4.5	158	11348-03-20G	789.99
8	Rating + 4.5	235	GV-R9070XTGAMING OC-16GD	810.00
9	Rating + 4.2	51	B580 CL 12GO	309.99
10	Rating + 4.6	75	RX7600 CL 8GO	299.99
11	Rating + 4.8	241	GV-N507TWF30C-16GD	1,169.99
12	Rating + 4.5	235	GV-R907GREGAMING OC-12GD	534.99
13	Rating + 4.6	166	RTX 5060 Ti 8G VENTUS 2X OC PLUS	509.99
14	Rating + 4.5	158	11348-01-20G	849.99
15	Rating + 5	1	RX-97TQICKBA	729.99
16	Rating + 4.6	393	RTX 5070 Ti 16G FRIEREN EDITION OC	1,149.99

17	Rating + 4.3	354	RADEON RX7900XTX 24G	1,029.99
18	Rating + 4.5	152	8346-00278	309.99
19	Rating + 4.6	166	RTX 5060 Ti 16G VENTUS 3X OC	799.99
20	Rating + 4.6	300	GV-N5080WF30C-16GD	1,599.99
21	Rating + 4.6	393	RTX 5070 Ti 16G VENTUS 3X OC BLACK	1,199.99
22	Rating + 4.7	27	RX-96TSW16BQ	499.99
23	Rating + 4.3	142	GV-N5070EAGLE OC-12GD	809.99
24	Rating + 4.5	248	RTX 5080 16G SHADOW 3X OC	1,579.99
25	Rating + 4.6	90	RX-97TRGBBB9	809.99
26	Rating + 4.8	158	GV-R906XGAMINGOCICE-16GD	549.99
27	Rating + 4.8	241	GV-N507TEAGLEOC ICE-16GD	1,119.99
28	Rating + 4.6	97	ROG-ASTRAL-RTX5090-032G-GAMING	4,829.99
29	Rating + 4.5	182	RX9070XT SL 16G	769.99
30	Rating + 4.3	142	GV-N5070WF30C-12GD	799.99
31	Rating + 4.8	158	GV-R9060XTGAMING OC-16GD	529.99
32	Rating + 4.7	210	RTX 5070 12G SHADOW 3X OC	809.99
33	Rating + 4.3	137	GV-N5090GAMING OC-32GD	4,699.99
34	Rating + 4.7	136	RTX 5090 32G GAMING TRIO OC	4,699.99
35	Rating + 4.6	248	PRIME-RTX5070TI-16G	1,119.99

	Shipping	Image_URL
0	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
1	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
2	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
3	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
4	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
5	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
6	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
7	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
8	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
9	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
10	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
11	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
12	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
13	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
14	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
15	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
16	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
17	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
18	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
19	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
20	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
21	None	https://c1.neweggimages.com/productimage/nb300...
22	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
23	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
24	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
25	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
26	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
27	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
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30	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
31	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
32	Free Shipping	https://c1.neweggimages.com/productimage/nb300...
33	Free Shipping	https://c1.neweggimages.com/productimage/nb300...

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34 $19.99 Shipping https://c1.neweggimages.com/productimage/nb300...
35   Free Shipping https://c1.neweggimages.com/productimage/nb300...
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In []: